

DOT PHMSA Pipeline Safety Regulatory Reference

49 CFR Part 191 — Transportation of Natural Gas by Pipeline

Incident Reporting Requirements & PHMSA Form 7100.1

1. Regulatory Authority & Applicability

1.1 Governing Regulations

- **49 CFR Part 191:** Transportation of Natural Gas by Pipeline — Annual Reports, Incident Reports, and Safety-Related Condition Reports
- **49 CFR Part 192:** Transportation of Natural and Other Gas by Pipeline — Minimum Federal Safety Standards
- **Administering agency:** U.S. Department of Transportation, Pipeline and Hazardous Materials Safety Administration (PHMSA)
- **Applicability:** All operators of natural gas transmission pipelines in interstate and intrastate commerce

1.2 Key Definitions (49 CFR 191.3)

- **Incident:** An event involving a gas pipeline facility that results in: (1) a death or personal injury requiring hospitalization; (2) estimated property damage of \$50,000 or more; (3) unintentional estimated gas loss of 3 million cubic feet or more; (4) an event that is significant in the judgment of the operator
 - **Significant incident:** Any event the operator determines warrants reporting based on potential consequences
 - **Transmission line:** A pipeline, other than a gathering line, that transports gas in interstate or intrastate commerce
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2. Incident Reporting Requirements (49 CFR 191.5)

2.1 Immediate Notification — 5-Minute Rule

An operator must give notice of each incident **as soon as practicable but no later than 1 hour** after confirmed detection to:

DOT PHMSA National Response Center (NRC) - Phone: **1-800-424-8802** (24 hours/day, 7 days/week) - Available: All hours, all days including holidays

Incidents requiring immediate notification: - Death or injury requiring hospitalization - Estimated property damage $\geq$ \$50,000 (including cost of gas lost) - Uncontrolled release of gas (any volume) that results in fire or explosion - Unintentional estimated gas loss $\geq$ 3 MMSCF - Emergency shutdown of pipeline facilities - Any event the operator determines is significant

2.2 Written Report — 30-Day Requirement

A written report on **PHMSA Form 7100.1** must be submitted within **30 days** of the incident to: - Online: PHMSA Portal at <https://portal.phmsa.dot.gov> - Mail: PHMSA, Information Resources Manager, PHP-10, 1200 New Jersey Ave SE, Washington, DC 20590

2.3 Supplemental Report

If new information becomes available after the initial report, a supplemental report must be filed within 30 days of obtaining the new information.

3. PHMSA Form 7100.1 — Required Fields

3.1 Section A: Operator Information

Field	Description	Example
Operator name	Legal name of pipeline operator	Midstream Gas Transmission LLC
Operator ID	PHMSA-assigned operator ID	31234
Operator address	Principal place of business	1234 Energy Blvd, Houston, TX 77002
Contact name	Person completing report	[Name], Pipeline Integrity Manager
Contact phone	Direct line	[Phone number]
Contact email	Direct email	[Email address]

3.2 Section B: Pipeline System Information

Field	Description	Example
Pipeline/facility name	Name of the pipeline system	Permian Basin Transmission System
Pipeline system ID	PHMSA system ID	12345
Commodity	Type of gas transported	Natural gas
Pipeline type	Transmission / gathering / distribution	Transmission
State(s)	State(s) where incident occurred	TX
County	County of incident	Midland County
Nearest city	Nearest populated area	Midland, TX
Mile post / location	Specific location on pipeline	Mile marker 118–134
Latitude / Longitude	GPS coordinates of incident	31.9974° N, 102.0779° W

3.3 Section C: Incident Information

Field	Description	Example
Incident date/time	When incident occurred (UTC)	2026-03-26 09:10 UTC
Incident type	Release / fire / explosion / other	Unintentional release
Cause category	Equipment failure / corrosion / excavation damage / other	Equipment failure — corrosion
Cause subcategory	Specific cause	External corrosion
Was incident a rupture?	Yes / No	No
Was there ignition?	Yes / No	No
Was there explosion?	Yes / No	No
Was there a fire?	Yes / No	No

3.4 Section D: Consequences

Field	Description	Example
Fatalities — employees	Number	0
Fatalities — contractors	Number	0
Fatalities — public	Number	0
Injuries — employees	Number requiring hospitalization	0
Injuries — contractors	Number requiring hospitalization	0
Injuries — public	Number requiring hospitalization	0

Field	Description	Example
Property damage — operator	Estimated USD	\$125,000
Property damage — other	Estimated USD	\$0
Volume of gas released	Estimated MCF	460 MCF
Did release reach water?	Yes / No	No
Did release reach HCA?	High Consequence Area — Yes / No	No

3.5 Section E: Pipeline Characteristics at Incident Location

Field	Description	Example
Pipe outside diameter	Inches	24 in
Wall thickness	Inches	0.375 in
Pipe material	Steel / plastic / other	Steel
Pipe specification	API 5L grade	X65
Year installed	Year of installation	1998
Seam type	ERW / DSAW / seamless	DSAW
Coating type	Fusion bonded epoxy / tape / other	Fusion bonded epoxy
Cathodic protection	Yes / No	Yes
MAOP	Maximum allowable operating pressure (psi)	850 psi
Operating pressure at time of incident	psi	742 psi
Depth of cover	Inches	36 in
Location class	1 / 2 / 3 / 4 (per 49 CFR 192.5)	1

3.6 Section F: Emergency Response

Field	Description	Example
Time operator notified	UTC	2026-03-26 09:14 UTC
Time NRC notified	UTC	2026-03-26 09:18 UTC
NRC report number	Assigned by NRC	NRC-2026-XXXXXX
Time pipeline isolated	UTC	2026-03-26 09:22 UTC
Isolation method	Valve closure / ESD / other	Remote valve closure (V-112, V-122)
Time flow stopped	UTC	2026-03-26 09:25 UTC
Emergency responders notified	Fire / police / LEPC / other	Local fire department, county emergency management
Evacuation required?	Yes / No	No

Field	Description	Example
Number evacuated	If yes	0

3.7 Section G: Cause Analysis

Field	Description
Primary cause	Operator's determination of root cause
Contributing factors	Secondary causes or conditions
Inspection history	Last ILI run date and results
Anomaly identified in prior inspection?	Yes / No — if yes, describe
Corrective action taken	Immediate repairs and long-term mitigation
Preventive measures	Steps to prevent recurrence

4. Leak Rate Thresholds & Reporting Triggers

4.1 Reporting Decision Matrix

Leak Rate	Estimated Volume	Reporting Required	Timeframe
Any rate with injury/death	Any	Yes — NRC + Form 7100.1	NRC within 1 hour; written within 30 days
Any rate with fire/explosion	Any	Yes — NRC + Form 7100.1	NRC within 1 hour; written within 30 days
>3 MMSCF total release	>3,000 MCF	Yes — NRC + Form 7100.1	NRC within 1 hour; written within 30 days
Property damage ≥\$50,000	Any	Yes — Form 7100.1	Written within 30 days
Operator-determined significant	Any	Yes — NRC + Form 7100.1	NRC within 1 hour; written within 30 days
Minor release, no consequences	<3 MMSCF, no damage	No federal report required	Document internally

4.2 Volume Conversion Reference

- 1 MMSCFD = 1,000 MCF/day = 1,000,000 SCF/day
- 3 MMSCF threshold = 3,000 MCF = 3,000,000 SCF
- At 0.5 MMSCFD leak rate: threshold reached in 6 days
- At 2.0 MMSCFD leak rate: threshold reached in 1.5 days

- At 5.0 MMSCFD leak rate: threshold reached in 14.4 hours
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5. Safety-Related Condition Reports (49 CFR 191.23)

5.1 When Required

An operator must submit a Safety-Related Condition Report within **5 days** of confirming any of the following: - General deterioration of pipeline that could lead to imminent hazard - Discovered defect that could cause failure if not corrected - Operating pressure exceeding MAOP - Unintended movement or abnormal loading of pipeline

5.2 Submission

- Form: PHMSA Form 7100.2
 - Submit to: PHMSA Office of Pipeline Safety, PHP-10
 - Also notify: State pipeline safety authority (Texas Railroad Commission for TX pipelines)
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6. Texas Railroad Commission (TxRRC) Requirements

6.1 State Reporting (in addition to federal)

Texas intrastate pipelines must also report to: - **Texas Railroad Commission, Pipeline Safety Division** - Phone: (512) 463-6838 - Online: <https://www.rrc.texas.gov/pipeline-safety/>

6.2 TxRRC Reporting Thresholds

- Same triggers as federal (49 CFR 191.5) plus:
- Any release that results in public evacuation
- Any release within 1,000 feet of a school, hospital, or public building
- Any release that results in media coverage

6.3 TxRRC Form P-18 (Incident Report)

Required within 30 days for all reportable incidents. Fields mirror PHMSA Form 7100.1 with additional Texas-specific fields for: - RRC pipeline permit number - RRC district office notification - Texas county and survey information

7. Draft Incident Notification Template

Use this template for the initial NRC phone notification:

PHMSA NATIONAL RESPONSE CENTER NOTIFICATION

Operator: [Company Name]
Operator ID: [PHMSA ID]
Pipeline system: [System Name]
Contact: [Name], [Title]
Contact phone: [Phone number]

Incident summary:

- Date/time of incident: [UTC]
- Location: Mile marker [XX], [County], [State]
- Coordinates: [Lat/Long]
- Incident type: Unintentional gas release
- Estimated release rate: [X.X] MMSCFD
- Estimated total release to date: [X] MCF
- Cause (preliminary): [Suspected cause]
- Injuries/fatalities: None / [Number]
- Fire or explosion: No / Yes
- Pipeline isolated: Yes – valves [V-XX] and [V-XX] closed at [UTC]
- Emergency responders notified: [Yes/No – agencies]
- Evacuation: None required / [Number] evacuated

Actions taken:

1. Pipeline isolated at [time]
2. NRC notified at [time]
3. Inspection crew dispatched to mile marker [XX]
4. [Additional actions]

NRC report number: [Assigned by NRC operator]

8. Post-Incident Requirements

8.1 Incident Investigation

Per 49 CFR 192.617, operators must: - Investigate each incident to determine cause
- Review procedures to determine if changes are needed - Implement corrective actions within 1 year - Document investigation findings and corrective actions

8.2 Return to Service

Before returning an isolated segment to service after a leak: 1. Repair or replace damaged pipe section 2. Pressure test repaired section to $1.25 \times$ MAOP for minimum 8 hours 3. Inspect repair with appropriate NDE (radiography or ultrasonic) 4. Document repair in pipeline records 5. Notify PHMSA of return to service if incident was reportable 6. Update ILI baseline if repair affects anomaly assessment

8.3 Record Retention

- Incident reports: retain for life of pipeline
- SCADA data for incident period: retain minimum 5 years
- Repair records: retain for life of pipeline
- Pressure test records: retain for life of pipeline